

Name: _____

Date: _____

DIVISION, (SHORT DIVISION)

LO: I can use short division (bus-stop) to divide by a single-digit number.

What is short division?

Short division is a quick written method for dividing by a **SINGLE**-digit number. We carry remainders mentally.

When to use: dividing by 2-9. For 10+, use long division.

Examples

●	$84 \div 4 = 21$	no remainder
●	$85 \div 4 = 21 \text{ r } 1$	with remainder
●	$936 \div 6 = 156$	3-digit
●	Carry to the LEFT	carry to the RIGHT
●	Skip a digit if too small	write 0 above

Key idea

Divide each digit left-to-right. If a digit is too small, carry to the next as a remainder.

$$\textcircled{936} \div \textcircled{6} = \textcircled{156}$$

Left to right; carry small remainders

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - no remainder

Calculate $84 \div 4$

$$8 \div 4 = 2 \text{ (write above 8)}$$

$$4 \div 4 = 1 \text{ (write above 4)}$$

Read off: 21

$$= 21$$

Check: $21 \times 4 = 84$

Example 2 - with carry

Calculate $75 \div 3$

$$7 \div 3 = 2 \text{ r } 1 \text{ (carry the 1 to next digit 15)}$$

$$15 \div 3 = 5$$

Read off: 25

$$= 25$$

The remainder '1' from $7 \div 3$ joins the next digit 5 to make 15.

Example 3 - with remainder

Calculate $85 \div 4$

$$8 \div 4 = 2$$

$$5 \div 4 = 1 \text{ r } 1$$

Read off: 21 r 1

$$= 21 \text{ r } 1$$

Final remainder is left over because no more digits to bring down.

Example 4 - writing zero

Calculate $612 \div 6$

$$6 \div 6 = 1$$

$$1 \div 6 = 0 \text{ r } 1 \text{ (carry 1 12)}$$

$$12 \div 6 = 2$$

$$= 102$$

When a digit doesn't divide, write 0 in the answer.

YOUR TURN – PRACTICE (deliberate practice)

1. Short division (no carry)

Each digit divides exactly.

- a) $48 \div 4$
- b) $96 \div 3$
- c) $84 \div 2$
- d) $636 \div 3$

2. With carry

Carry small remainders to the next digit.

- a) $75 \div 3$
- b) $92 \div 4$
- c) $68 \div 4$
- d) $238 \div 7$

3. With remainder

Write 'quotient r remainder'.

- a) $85 \div 4$
- b) $47 \div 5$
- c) $100 \div 7$
- d) $153 \div 6$

4. Writing zero in answer

If a digit doesn't divide, write 0 in the answer.

- a) $612 \div 6$
- b) $509 \div 5$
- c) $303 \div 3$
- d) $808 \div 4$

5. Mixed short division

Choose carefully and check.

- a) $744 \div 8$
- b) $925 \div 5$
- c) $486 \div 9$
- d) $327 \div 3$
- e) $1001 \div 7$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Carry remainders to the next digit on the RIGHT ☐

Correct answer: _____

Reason: _____

b) $612 \div 6 = 12$ (skip the 1) ☐

Correct answer: _____

Reason: _____

c) $85 \div 4 = 21 \text{ r } 1$ ☐

Correct answer: _____

Reason: _____

d) Short division works for any divisor ☐

Correct answer: _____

Reason: _____

e) Always check: quotient $\times$ divisor + remainder = dividend ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) **936 sweets** are shared equally between **8 children**. How many sweets each? Are any left over? Show your short division.

Expression: _____

Simplified: _____

b) A school orders **525 textbooks** packed in boxes of **7**. How many full boxes? Now imagine **530 books**. How many full boxes and how many spare?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

